

MAE5005 / MAE403 Computational Fluid Dynamics, Spring 2026 Computer Project

Overall notes: Two of you will work as a group. The contributions of each student in the group should be coordinated and described in your group report. Your group report must be typed up neatly using Word, LaTeX, or similar tool. Plots should be done properly using dimensionless quantities, at *professional quality*.

Schedule and grading:

1. Oral presentation and report for Part I: Monday, May 25, 10 minutes / group.
2. Oral presentation and report for Part II: TBD, 10 minutes / group.

I. Solving the linear advection equation with discontinuous initial condition in a periodic domain.

Consider the 1D linear advection equation

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0 \quad (1)$$

with $a = 1$, in a *periodic domain*, $-1 \leq x \leq 1$, meaning $u(x = -1, t) = u(x = 1, t)$ at all times. The initial condition is

$$u(x, t = 0) = \begin{cases} 1.0 & \text{if } -0.5 < x < 0.5 \\ 0.0 & \text{otherwise.} \end{cases} \quad (2)$$

In this case the analytical solution of the PDE is known, at all times. Now consider the following scheme provided in the lecture, based on the cell-wise linear reconstruction

$$u_j^{n+1} = u_j^n - \frac{a\Delta t}{\Delta x} (u_j^n - u_{j-1}^n) - \frac{a\Delta t}{2\Delta x} (\Delta x - a\Delta t) (\sigma_j^n - \sigma_{j-1}^n) \quad (3)$$

where σ_j^n is a properly defined slope for cell j .

1. Assume $N = 160$ and $x_i = (i - 0.5)\Delta x$, and design properly the CFL number. Write three Fortran Codes using, respectively, the downwind slope (*i.e.*, the Lax-Wendroff scheme), the van Leer limiter (form 2 in the lecture), and the SUPERBEE limiter. Plot all solutions (three numerical solutions plus the analytical solutions if available) together at $t = 2.0$ and $t = 8.0$, respectively (one plot for each time). Support your numerical results with any analysis (in particular for the Lax-Wendroff scheme) you have learned in this class.
2. Comment on the results from the above three treatments. Also plot and compare the total variation (TV) as a function of time for the three cases (plot all TV data on a single figure). The total variation is defined as

$$TV = \sum_{\text{all } i} |u_{i+1} - u_i|. \quad (4)$$

3. Study the order of accuracy of the three schemes in Part 1. Explain how you design the details for obtaining the order of accuracy.
4. Study and compare the conditions for numerical stability for the three schemes in Part 2. Explain how you design the details for obtaining the condition for numerical stability.
5. *Open-ended question:* Propose one idea to improve the Lax-Wendroff scheme and implement your idea, to demonstrate that your idea actually improves the numerical results of the scheme.

II. Study of Two-Dimensional Cavity Flow. Download and understand the 2D lid-driven cavity flow code, under “9a. *Finite difference code for 2D cavity flow - AB*” in the Blackboard Class Fortran Demo Codes folder.

Performing the following components:

1. State the physical problem that the code is designed to solve, the relevant governing PDEs, boundary and initial conditions, and ALL the governing parameters.
2. De-code the spatial discretization schemes used in the code for the nonlinear term (*e.g.*, subroutine `hxcalc`) and the viscous term (*e.g.*, subroutine `axycalc`), respectively. Work out the *leading-order* truncation error terms for the nonlinear term and the viscous term, respectively.
3. Analyze how the boundary conditions are handled in the code. In your opinion, what are the orders of accuracy of various boundary-condition related treatments in the code.
4. Run the code for $Re = 100$, $Re = 1000$, and $Re = 5000$ with proper grid resolution and time step size, respectively, till the flow reaches the steady state. Compare the time evolution of velocity components at the center of the domain. Comment on the time required for the flow to reach the steady state. How would you scale this transient time based on the physics of the flow problem?
5. For the same three runs, compare the streamlines at the steady state by doing contour plots for the streamfunction. Also compare the velocity profiles at $x/H = 0.5$ (vertical cut in the middle of the domain) by actually plotting both the u and v velocity profiles. Comment on your results in physical terms. Compare your results directly with the data in AbdelMigid et al. (2017, Alexandria Engineering Journal 56, 123–135) whenever possible.
6. For the case of $Re = 1000$, compute the stream function and vorticity at the center of primary vortex (explain how you would obtain them in details), and then compare your results directly with the data in Kim and Moin (1985, J. Comp. Phys. 59: 308–323) and AbdelMigid et al. (2017, Alexandria Engineering Journal 56, 123–135)

7. For the case of $Re = 1000$ and grid resolution of 96×96 , run the code multiple times with different time step sizes. At what CFL number will the code become numerically unstable?
8. *Open-ended problem:* Propose one idea to improve the code and implement your idea, to demonstrate that your idea actually improves the numerical results.

Your group should hand in a report of what you have done and the simulations that you have conducted to address the different components. Your report should include a printout of your main code and relevant plots. Make sure that you describe the key physical (dimensionless) governing parameters and relevant numerical parameters for each case.